

Abhishek Bhadre

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SUMMARY

AI Engineer and M.S. Artificial Intelligence candidate with production experience building end-to-end ML systems — from computer vision pipelines deployed on Kubernetes to full-stack RAG applications serving LLM-powered responses. Strong Python and systems fundamentals, with a track record of shipping models that move real metrics: **+30%** detection accuracy in production, **99.2% F1** on text classification, and **25%** faster inference on deployed services. Passionate about designing scalable, reliable AI software that users actually touch.

EDUCATION

Illinois Institute of Technology — M.S., Artificial Intelligence
Chicago, IL | GPA: 3.0/4.0

Aug 2024 – May 2026

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Structures & Algorithms, Distributed Systems

Vishwakarma Institute of Technology — B.Tech., Instrumentation Engineering
Pune, India | GPA: 3.32/4.0

Aug 2019 – Jun 2023

EXPERIENCE

RAWATTECH — Data Science Engineer
Pune, India

Sep 2022 – May 2023

- Designed and shipped a YOLO-based computer vision pipeline in Python (OOP) that detects and classifies industrial pipe inventory from warehouse imagery, improving detection accuracy by **30%** and replacing a manual counting workflow.
- Containerized model inference with Docker and deployed on Kubernetes clusters, enabling horizontal scaling and cutting per-image inference latency in a production environment.
- Scaled training data from a few hundred to **10,000+** curated images; re-annotated **500+** edge cases and built a NumPy/Pandas augmentation pipeline that pushed model accuracy to **85.4% (+20% lift)** on held-out test data.
- Designed normalized SQL schemas to store image metadata and inference outputs, giving downstream teams a queryable source of truth for model performance and inventory analytics.

PROJECTS

CodeSyntax AI — Full-Stack RAG Developer Assistant | Python, FastAPI, Next.js, LangChain, Pinecone, Docker, GitHub Actions

- Built and deployed a production Retrieval-Augmented Generation application that answers Python developer questions by retrieving from Python Docs, RealPython, and StackOverflow and generating grounded responses with GPT-4o-mini and OpenAI embeddings stored in a Pinecone vector index.
- Engineered an LLM-based intent router with query classification, authority-weighted re-ranking, and multi-turn conversation support via chat-history condensing in LangChain — reducing irrelevant retrievals and improving answer quality.
- Wrote **35+** pytest unit tests and a **55-question** evaluation harness to regression-test answer quality; set up a GitHub Actions CI/CD pipeline with Docker builds deploying to Railway (API) and Vercel (frontend).
- Shipped a responsive React 19 + TypeScript + Tailwind chat UI with markdown rendering, syntax highlighting, and inline source citations.

AuthentiText — AI-Generated Text Detection Engine | Python, XGBoost, DistilBERT, HuggingFace, GCP, Docker

- Built a dual-model ensemble (XGBoost + fine-tuned DistilBERT) that detects AI-generated text with **99.2% F1** and **99.96% AUROC**, fusing **21** handcrafted linguistic features (perplexity, burstiness, lexical density) with transformer embeddings.
- Engineered the full NLP pipeline end-to-end: data collection via GPT-4 API and the HC3 dataset, feature extraction, hyperparameter tuning, and production deployment on GCP via Docker and GitHub Actions CI/CD.
- Benchmarked **4** model architectures (Logistic Regression, Random Forest, XGBoost, DistilBERT) with ROC curves, confusion matrices, and SHAP analysis; built a token-level heatmap visualizer for model explainability.

Amazon Review Sentiment Prediction | Python, Scikit-learn, TensorFlow, NLP

- Processed **32M+** Amazon reviews across **5** sentiment classes; trained and tuned **5** ML/DL models, improving accuracy by **20%** through hyperparameter optimization.
- Shipped a user-facing interface with real-time inference and confidence scoring, delivering **25%** faster predictions than the baseline notebook workflow.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Go, Java (familiar), C++ (familiar)

ML / AI: PyTorch, TensorFlow, Scikit-learn, XGBoost, HuggingFace Transformers, LangChain, RAG, LLMs, Semantic Search, Prompt Engineering, Computer Vision, NLP

Systems & Infra: Docker, Kubernetes, CI/CD (GitHub Actions), AWS, GCP, FastAPI, REST APIs, MySQL, Pinecone, FAISS, Git

Frontend: React, Next.js, Tailwind CSS

CERTIFICATIONS & ACHIEVEMENTS

- Microsoft Certified:** Azure Fundamentals (AZ-900)
- HackerRank:** SQL Problem Solving (2022)
- INGENIOUS 2021:** National Virtual Hackathon Participant